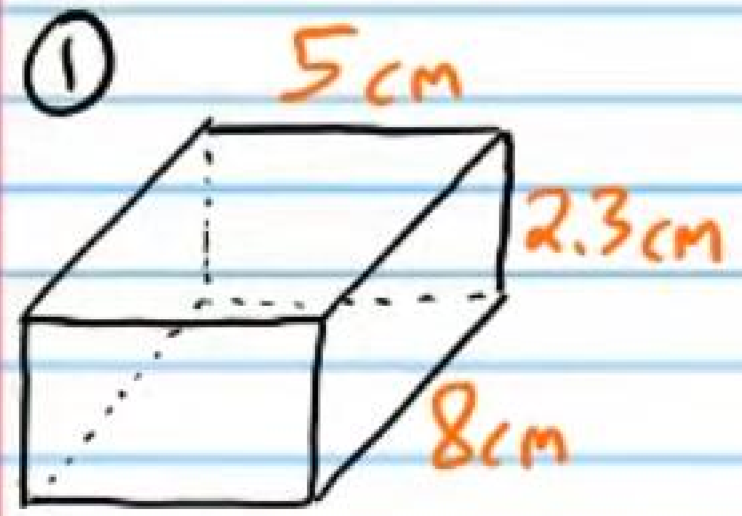
Name
Date
Course

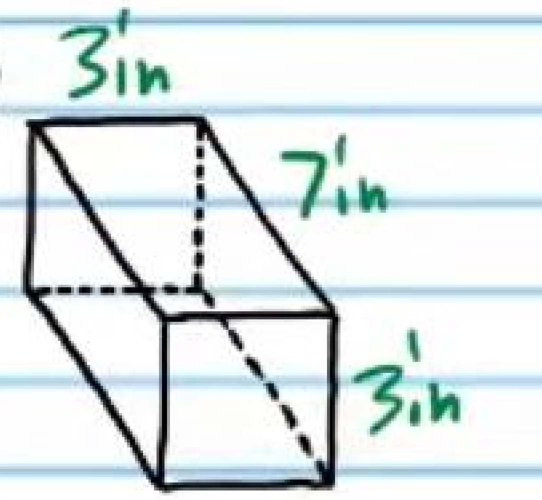
Volume (Homework)

Find the volume of each rectangular prism below.

Answers

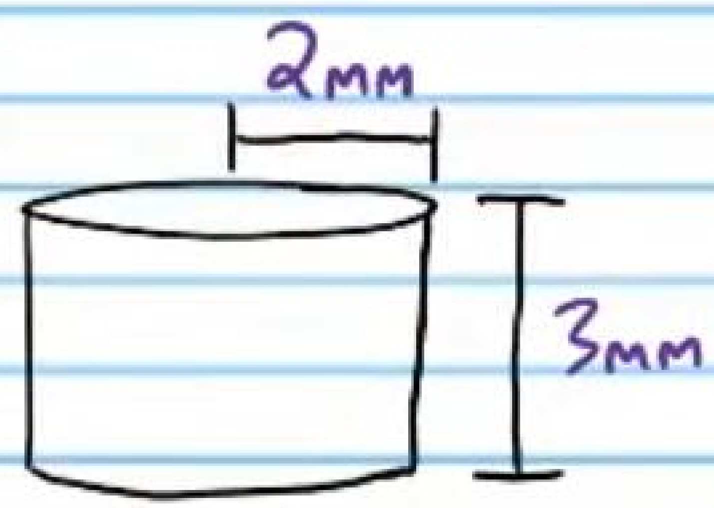
①


$\boxed{92\ cm^3}$

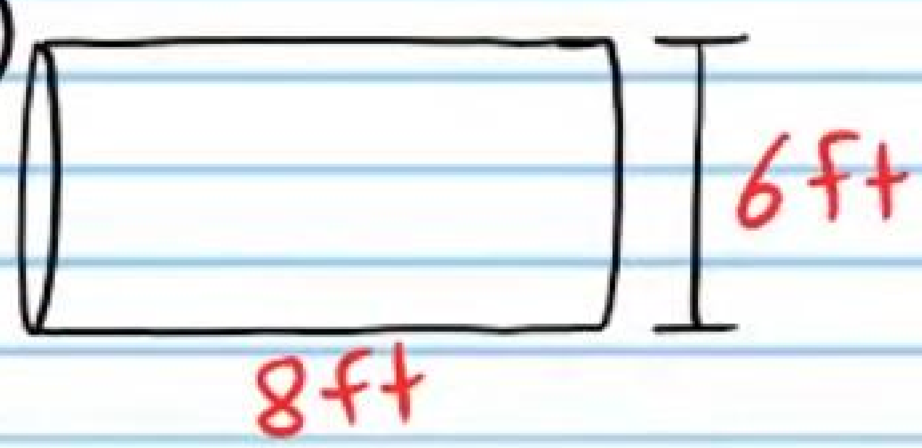
②


$\boxed{63\ in^3}$

Find the volume of each cylinder below.
Leave each answer as a multiple of π.

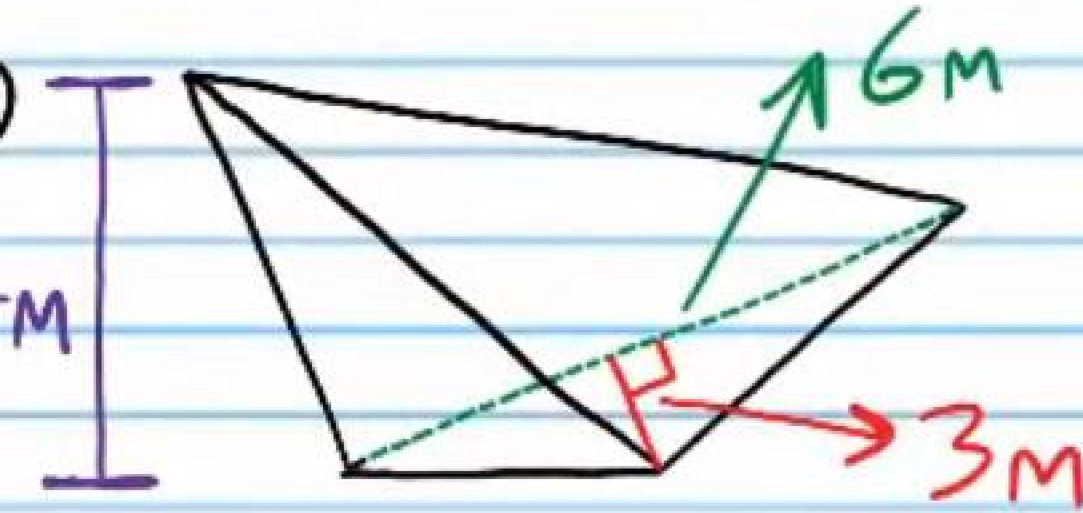
③


$\boxed{12\pi\ mm^3}$

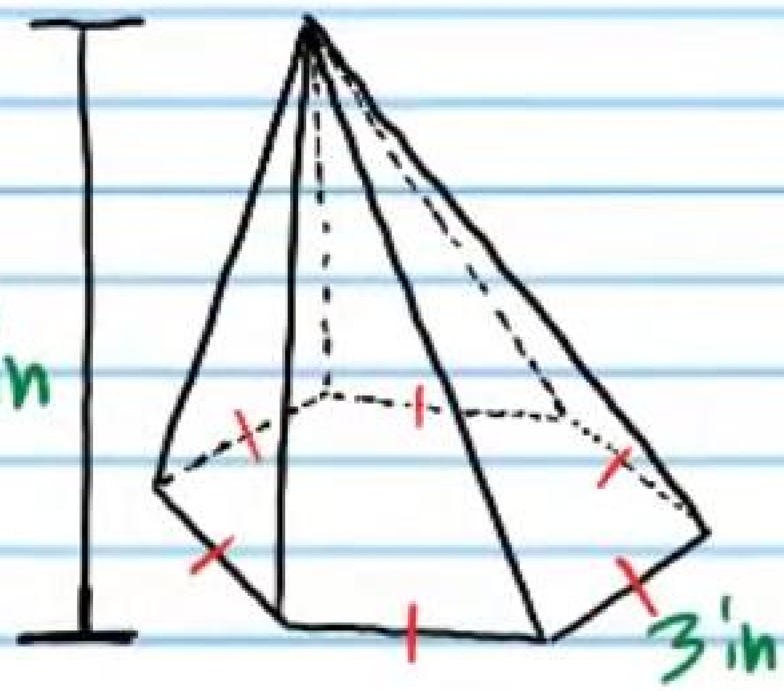
④


$\boxed{72\pi\ ft^3}$

Find the volume of each pyramid below. Round to the nearest tenth.

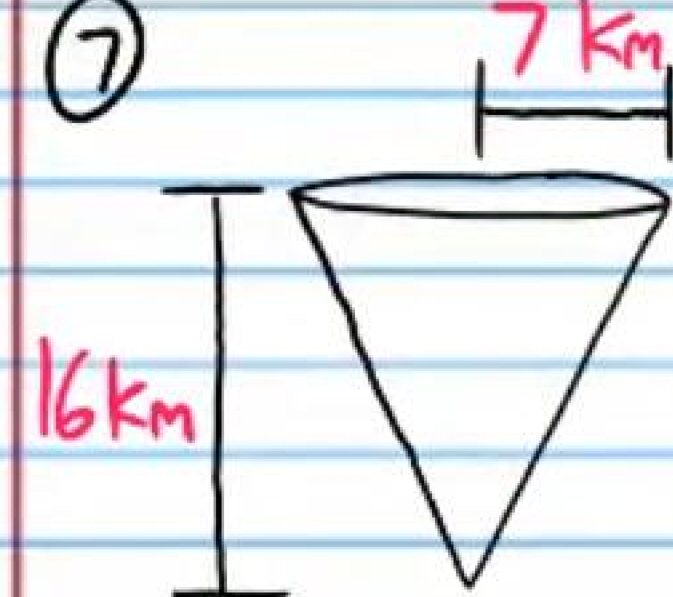
⑤


$\boxed{12\ m^3}$

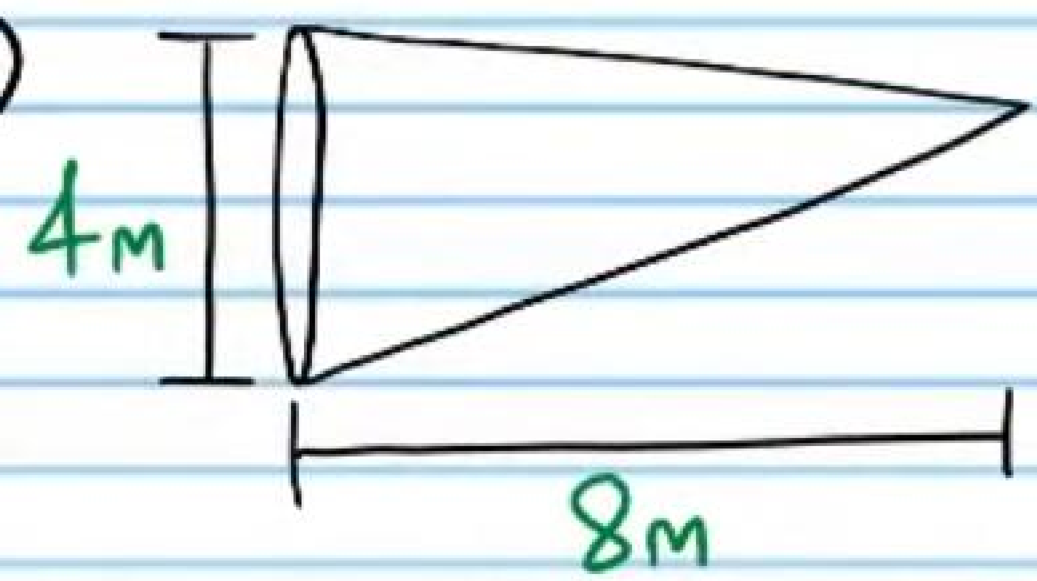
⑥


$\boxed{70.1\ in^3}$

Find the volume of each cone below. Round to the nearest hundredth.

⑦


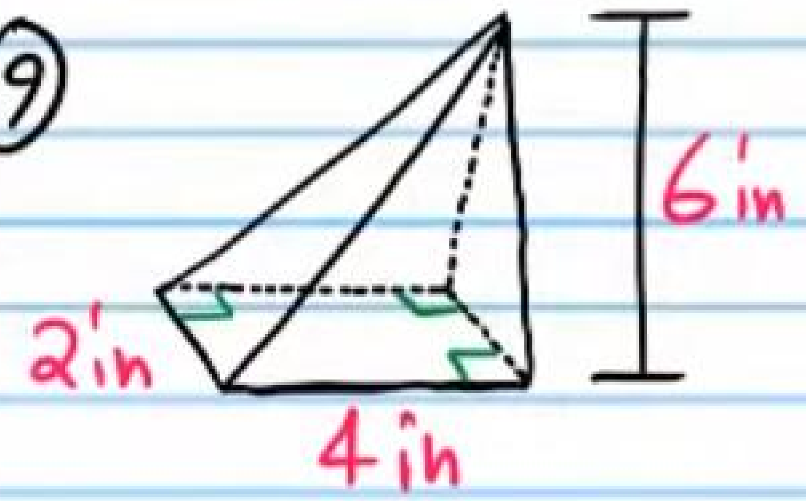
$\boxed{821\ km^3}$

⑧


$\boxed{33.51\ m^3}$

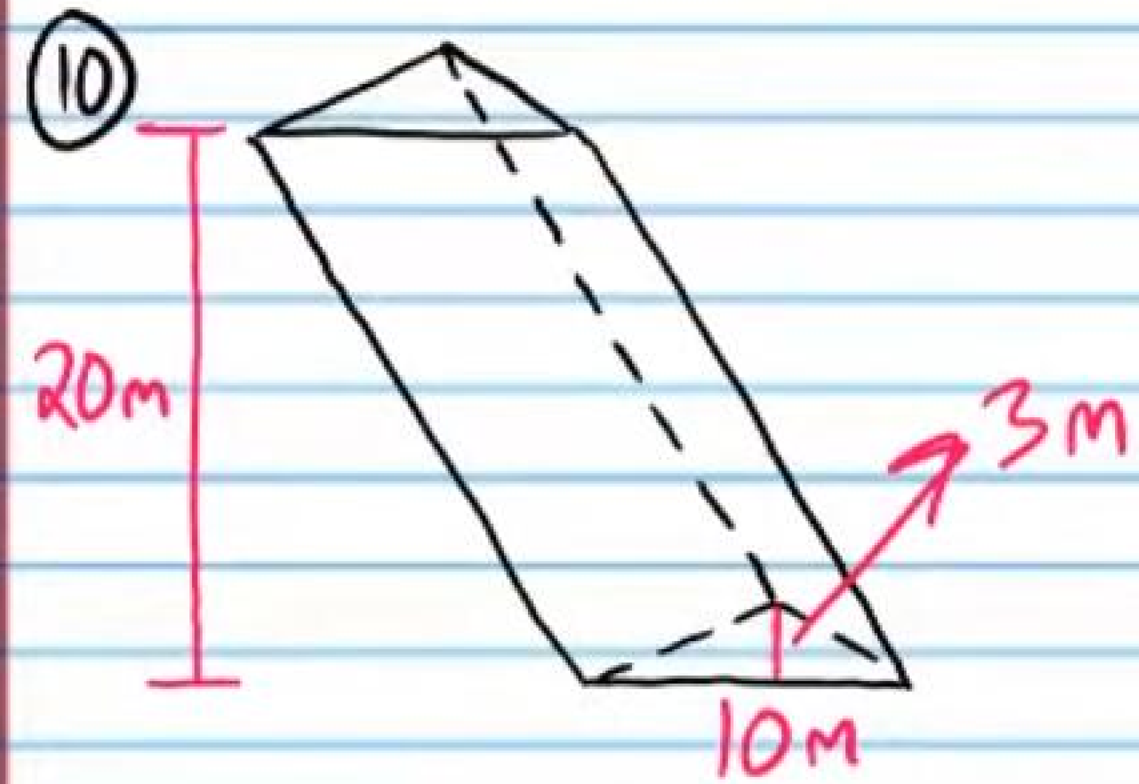
Find the volume of each figure below. For problem #11, leave your answer as a multiple of π.

(9)



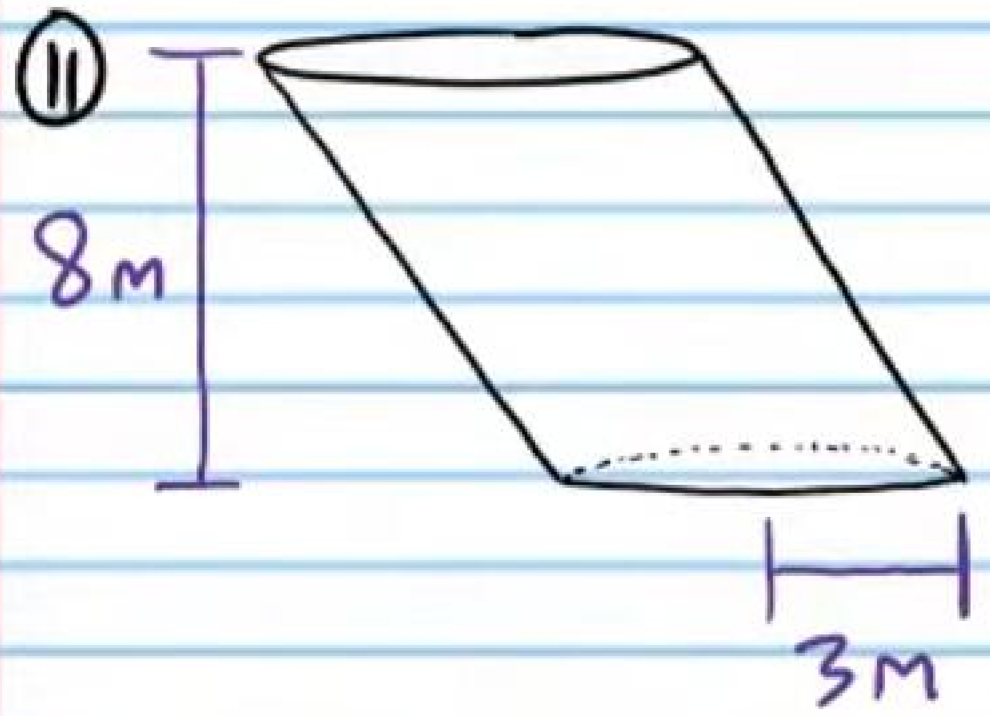
$\boxed{16\ in^3}$

(10)



$\boxed{300\ m^3}$

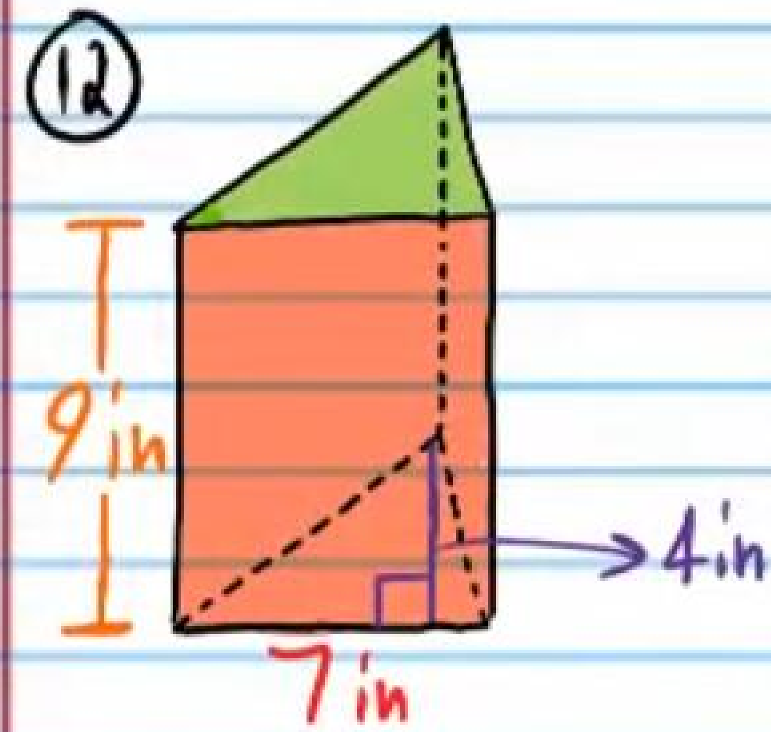
(11)



$\boxed{72\pi\ m^3}$

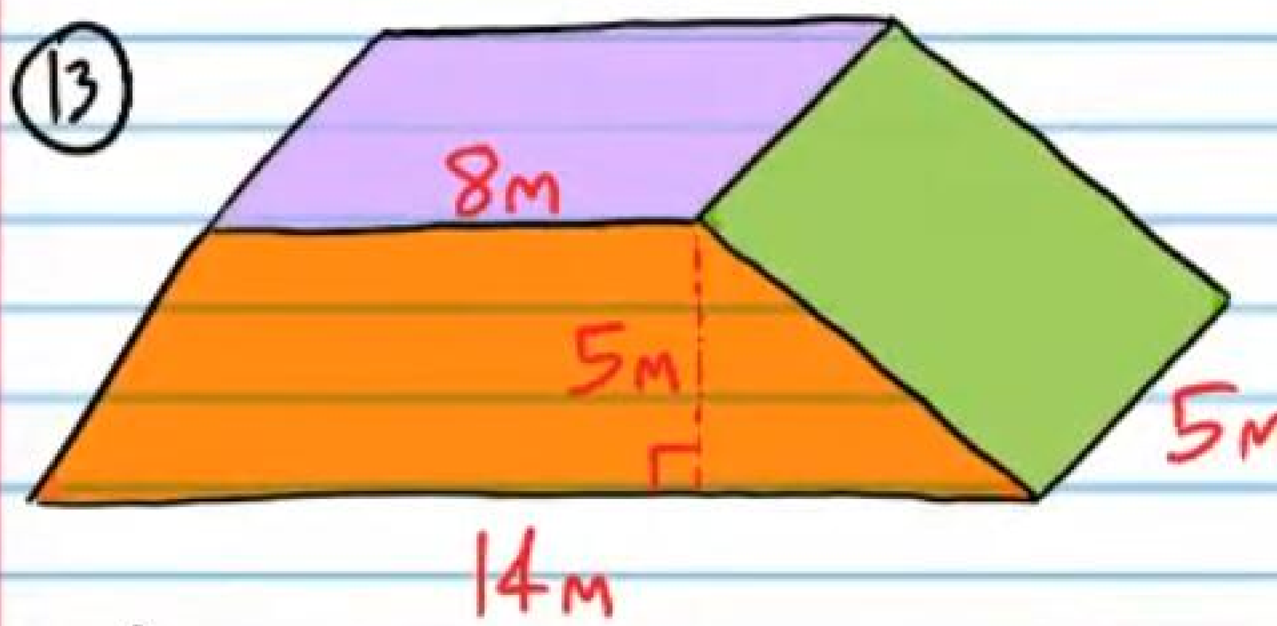
Find the volume of each prism below.

(12)



$\boxed{126\ in^3}$

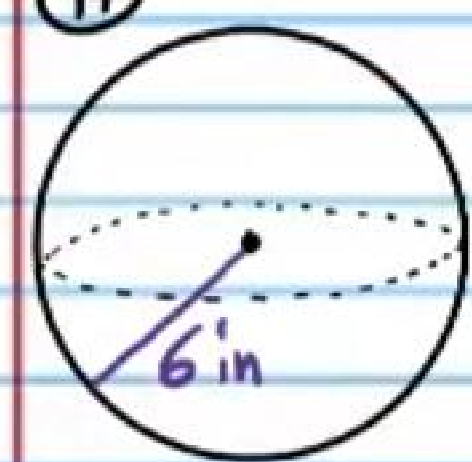
(13)



$\boxed{275\ m^3}$

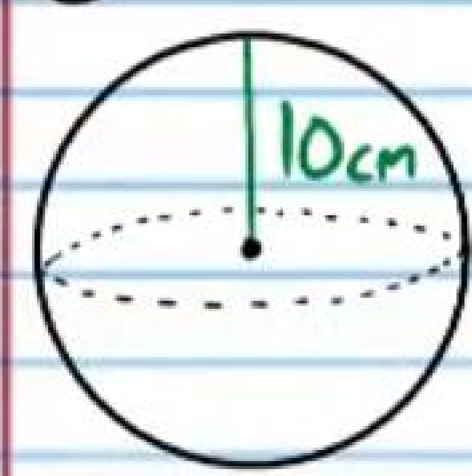
Find the volume of each sphere below. Round to the nearest tenth.

(14)



$\boxed{904.8\ in^3}$

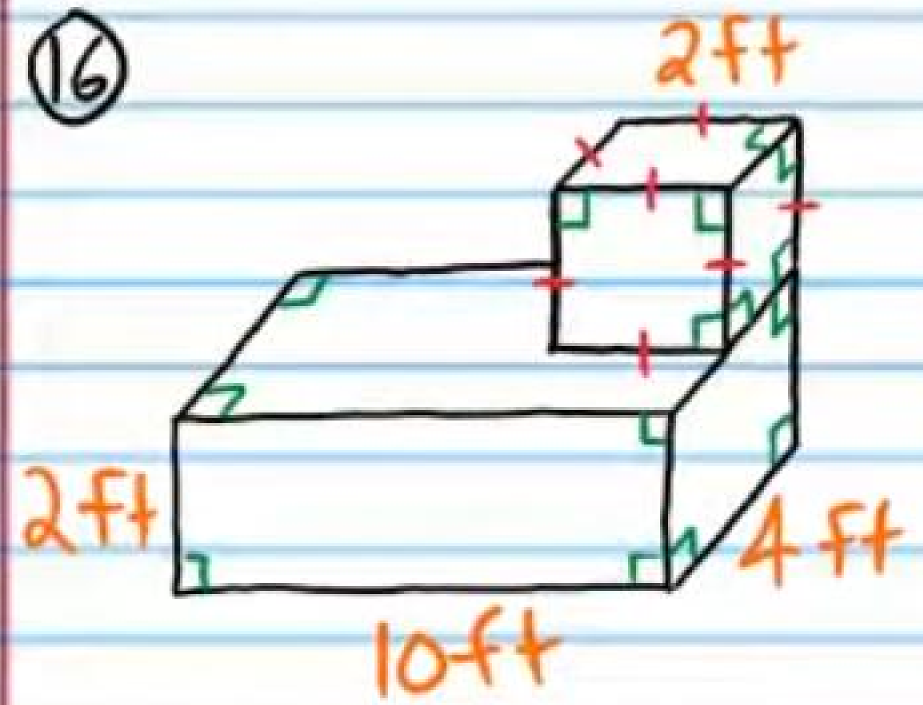
(15)



$\boxed{4{,}188.8\ cm^3}$

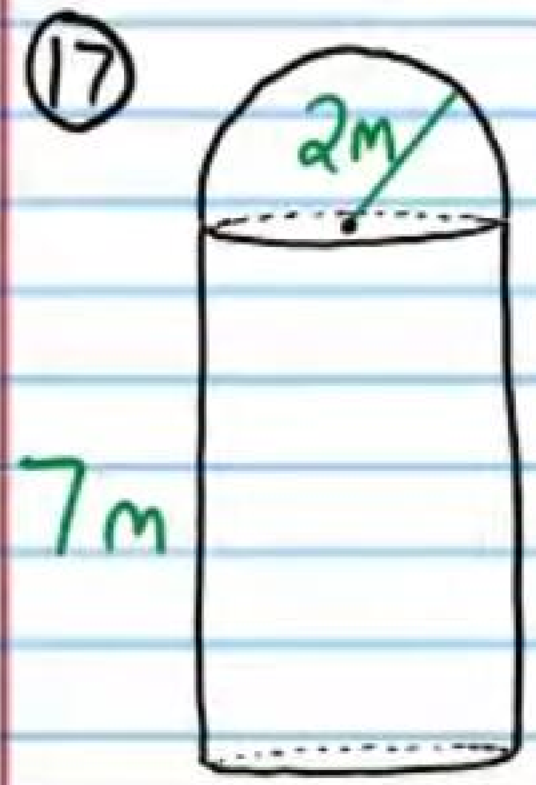
Find the volume of the following solids. Round to the nearest hundredth if necessary.

16



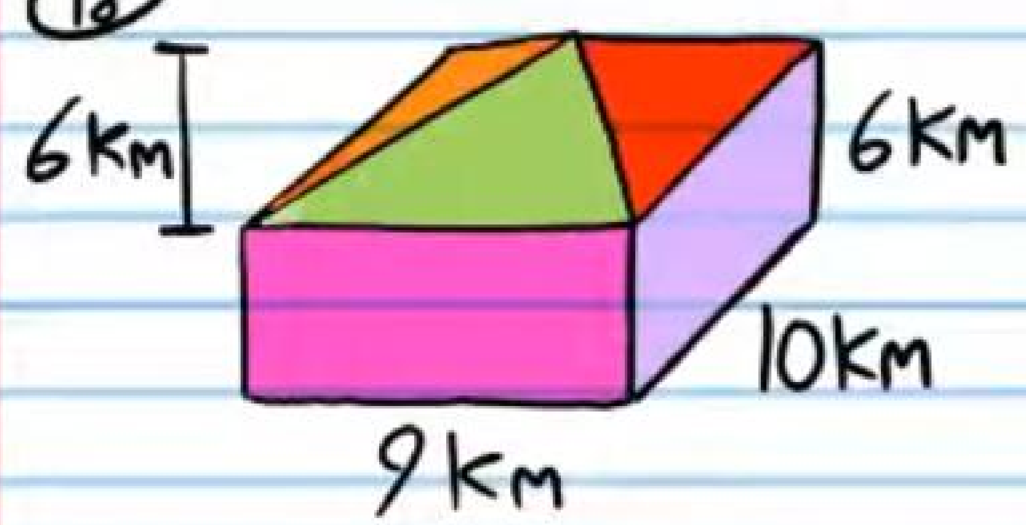
$\boxed{88\ ft^3}$

17



$\boxed{104.72\ m^3}$

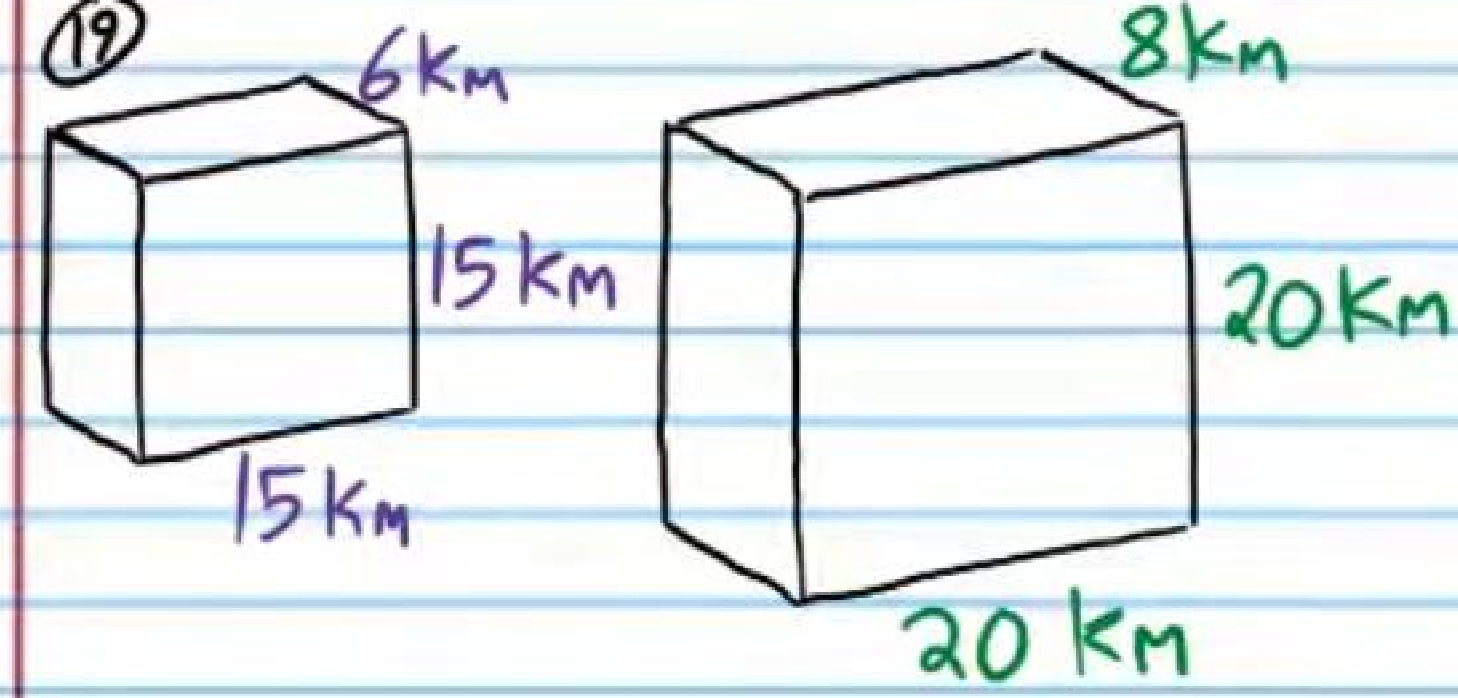
18



$\boxed{720\ km^3}$

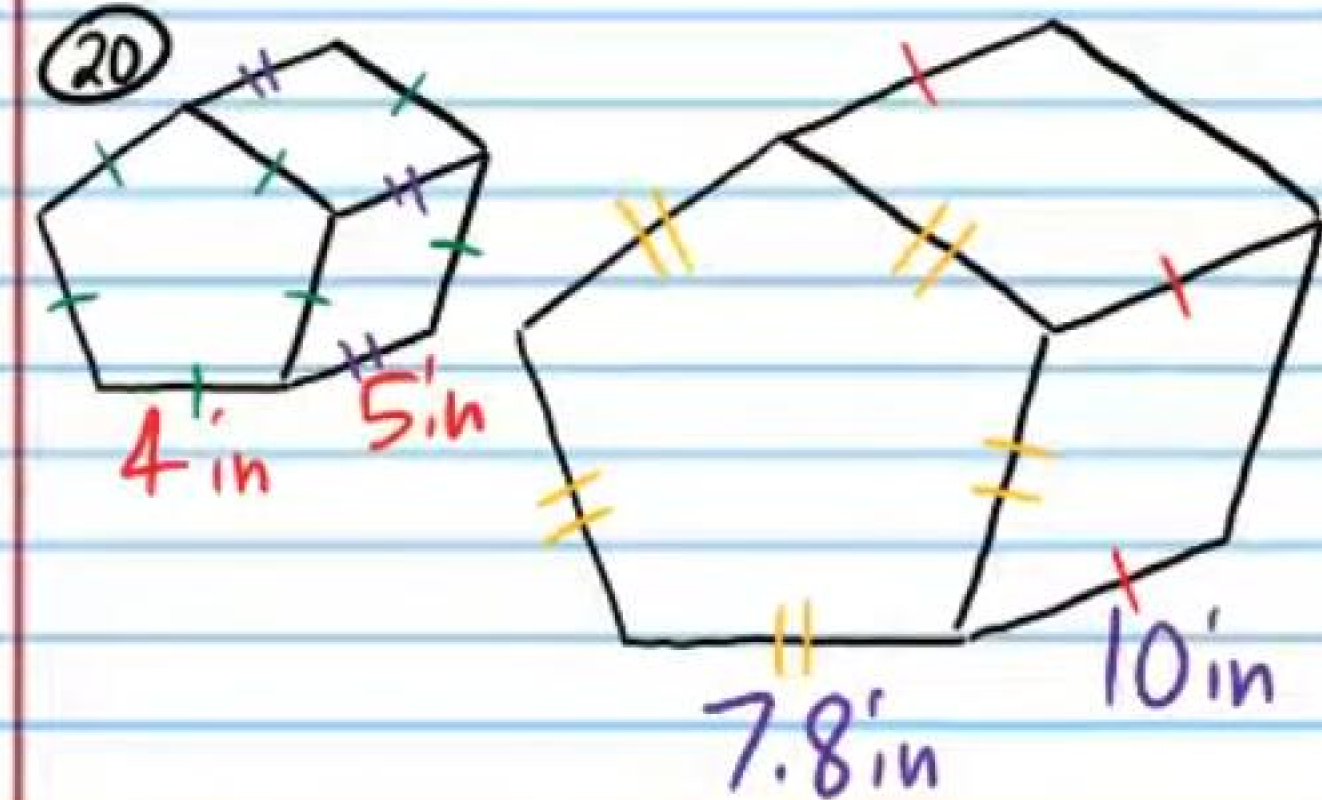
Determine if each pair of figures below contains similar solids.

19



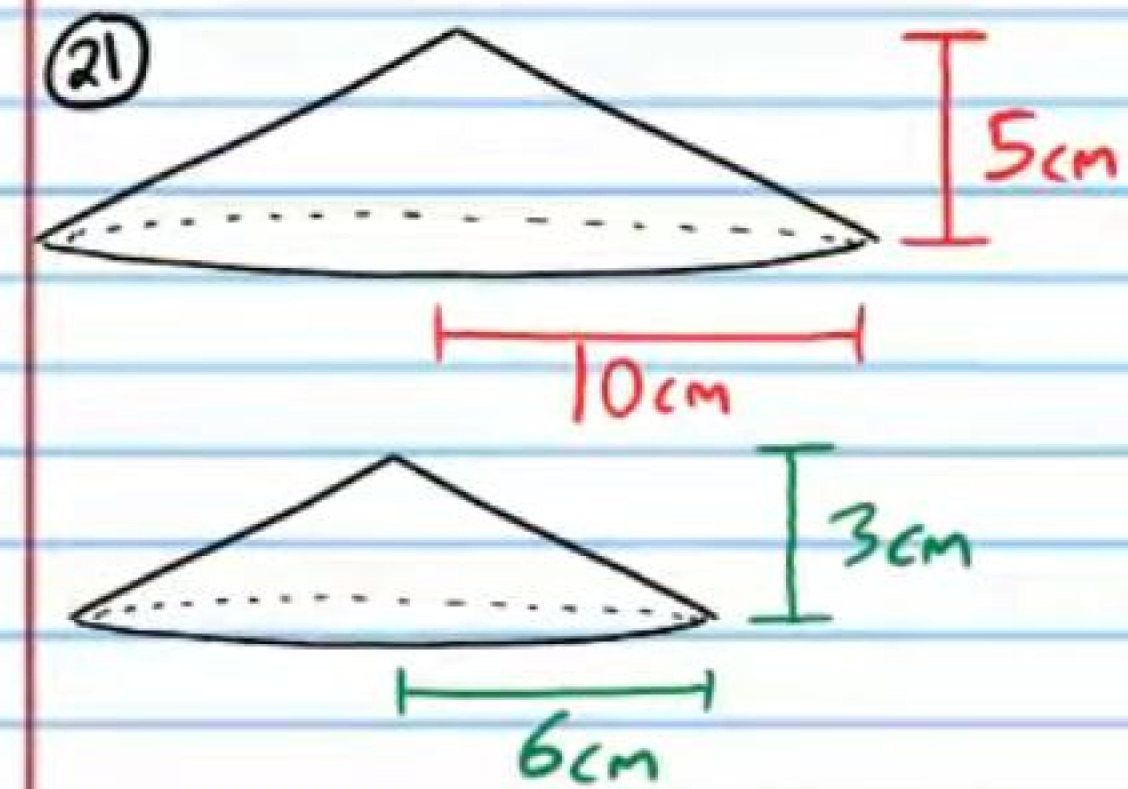
Yes

20



No

21



Yes

The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the Volume of the smaller solid, find the volume of the larger solid.

(22) $SF = \frac{5}{2}$, $V = 3\,cm^3$ $\boxed{46.875\,cm^3}$

(23) $SF = 3$, $V = 5\,ft^3$ $\boxed{135\,ft^3}$

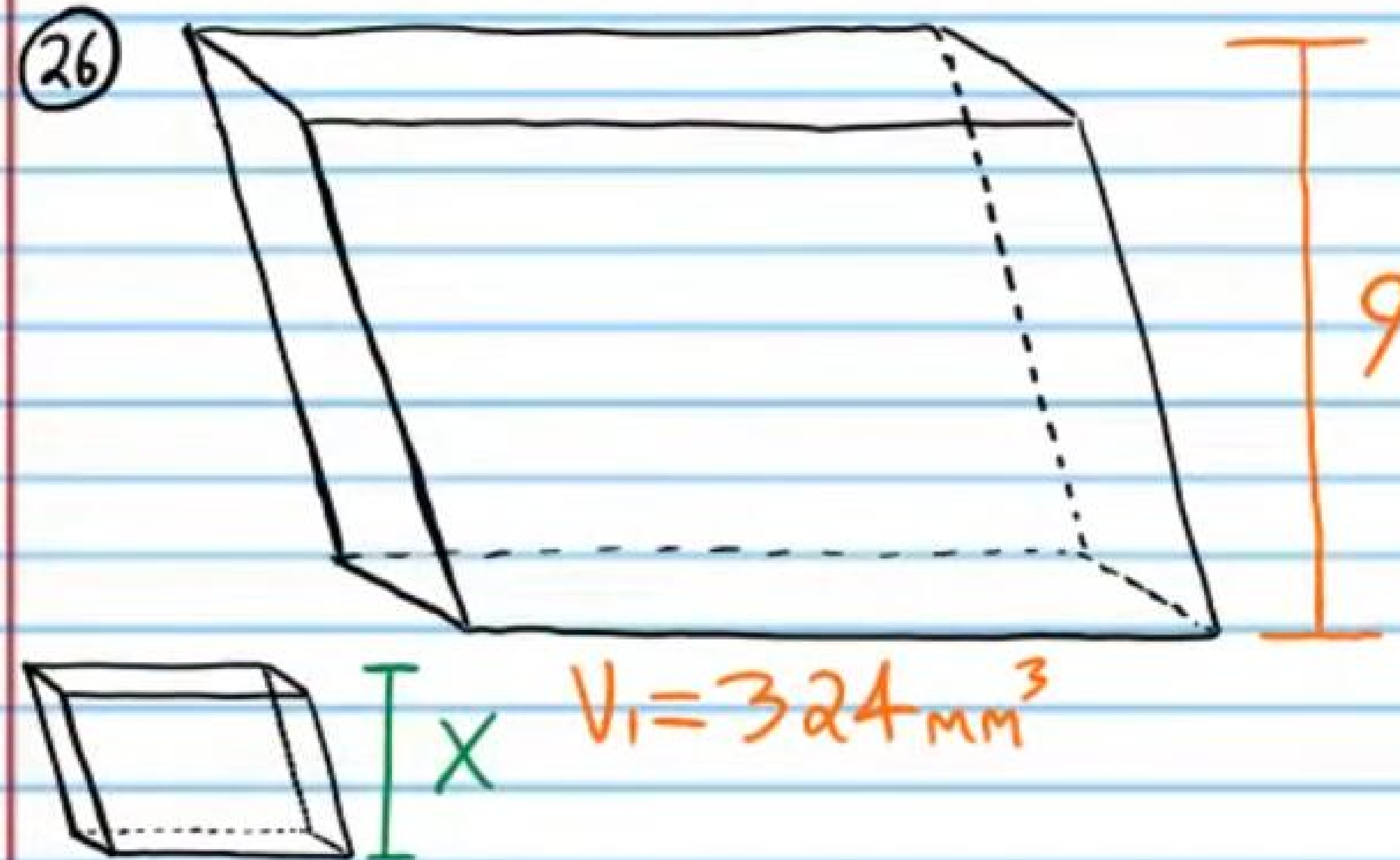
The ratio of corresponding sides (i.e. Scale factor) of two similar solids is written below. Given the Volume of the larger solid, find the volume of the smaller solid.

(24) $SF = \frac{1}{5}$, $V = 1\,mm^3$ $\boxed{1/125\,mm^3}$

(25) $SF = 6$, $V = 2\,in^3$ $\boxed{1/108\,in^3}$

If the following pairs of figures below are similar solids, find the unknown values.

(26)

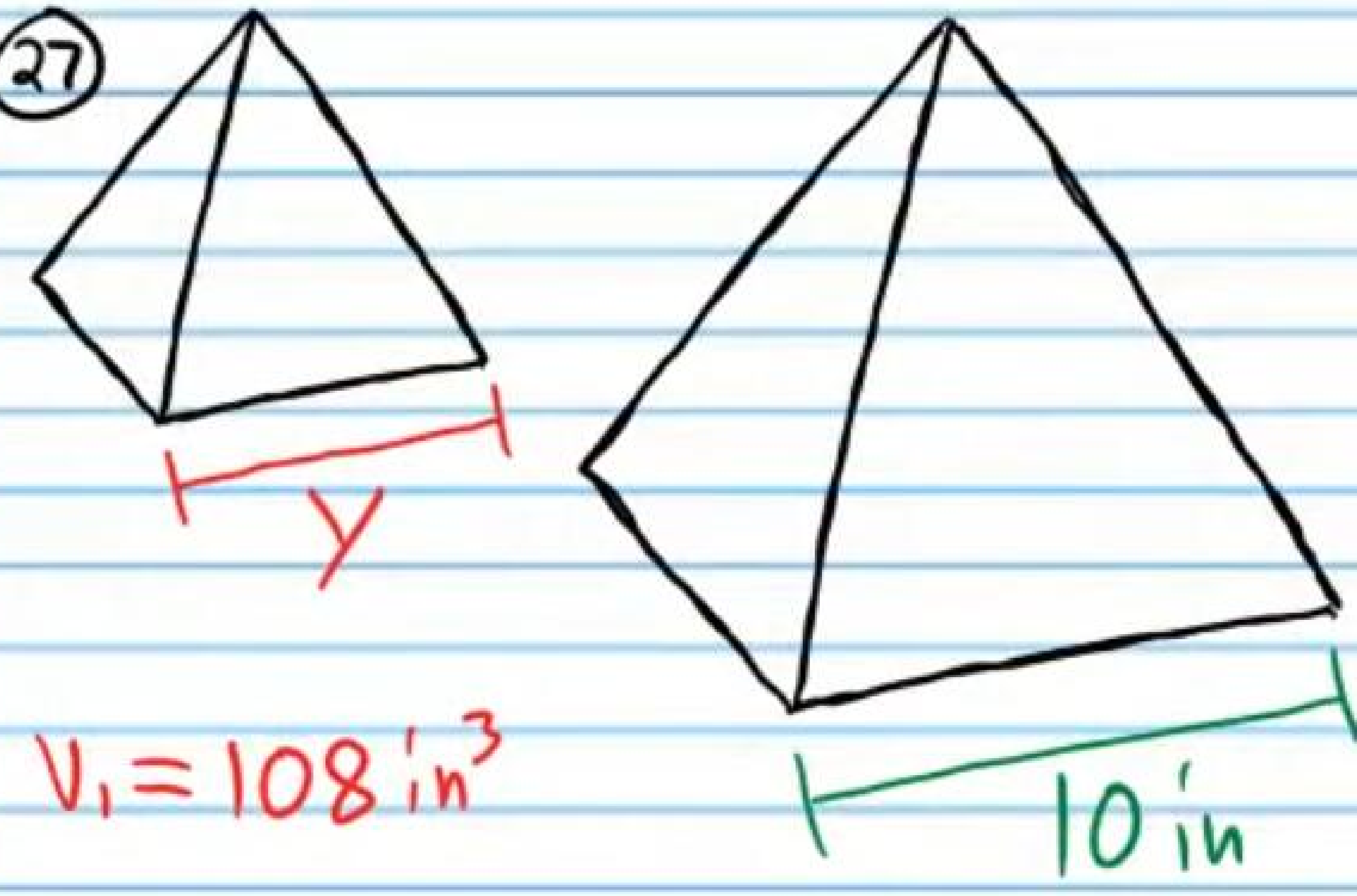


$V_1 = 324\,mm^3$

$V_2 = 12\,mm^3$

$\boxed{3\,mm}$

(27)



$V_1 = 108\,in^3$

$V_2 = 500\,in^3$

$\boxed{6\,in}$